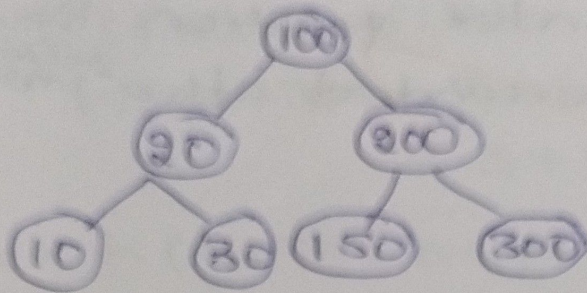


CO3-AT3 - Tree Traversal challenge

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Reg: 19252515

1)

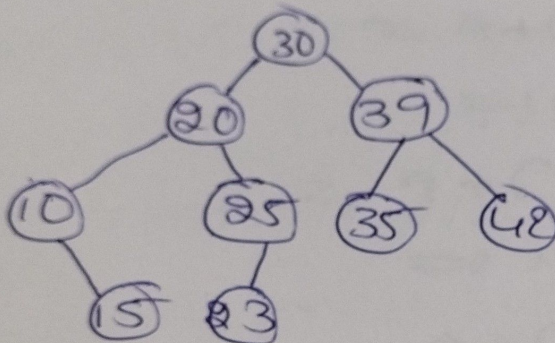


Preorder: 100, 20, 10, 30, 200, 150, 300 (NLR)

In order: 10, 20, 30, 100, 150, 200, 300 (LNR)

Post order: 10, 30, 20, 150, 300, 200, 100 (LRN)

2)

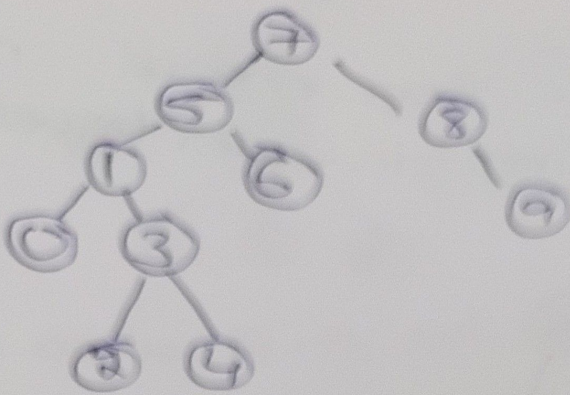


Inorder (LNR): 10, 15, 20, 23, 25, 30, 35, 39, 42

Pre order (NLR): 30, 20, 10, 15, 25, 23, 39, 35, 42

Post order (LRN): 15, 10, 23, 25, 20, 35, 42, 39, 30.

3) Assume the digits 7, 5, 1, 8, 3, 6, 0, 9, 2, are added in sequence into a binary search tree that is initially empty. what is the resulting tree's traversal sequence

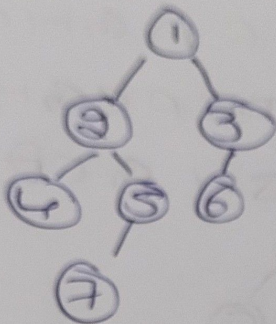


Preorder (NLP): 7, 5, 1, 0, 3, 2, 4, 6, 8, 9

Inorder (LNR): 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

postorder (LRN): 0, 2, 4, 3, 1, 6, 5, 9, 8, 7

5)

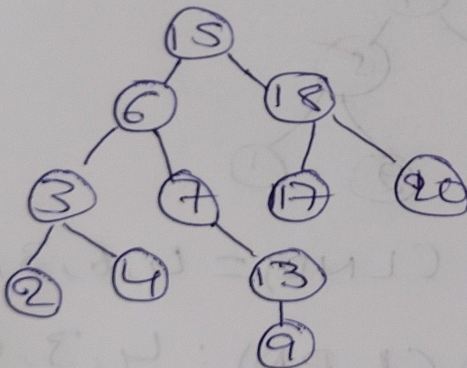


Inorder (LNR): 4, 2, 7, 5, 1, 3, 6

preorder (NLP): 1, 2, 4, 5, 7, 3, 6

postorder (LRN): 4, 7, 5, 2, 6, 3, 1

6)



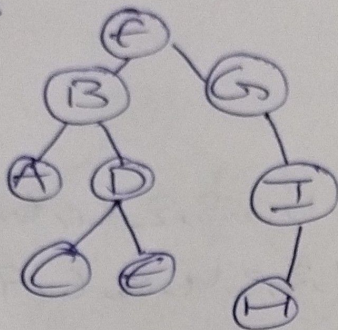
Inorder: 2, 3, 4, 6, 7, 9, 13, 15, 17, 18, 20

preorder: 15, 6, 3, 2, 4, 7, 13, 9, 18, 17, 20

Post order: 2, 4, 3, 9, 13, 7, 6, 17, 20, 18, 15

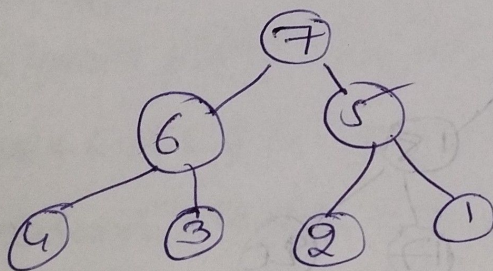
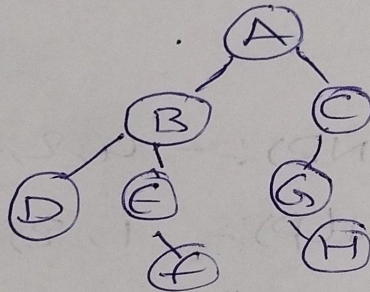
(LRN)

7) what is the sequence of nodes when applying in-order traversal on the binary



In order (LNR): A, C, D, E, B, F, G, H

8) In order - D B E F A G H C
 pre order - A B D E F C G H
 post order D E F B H G C A

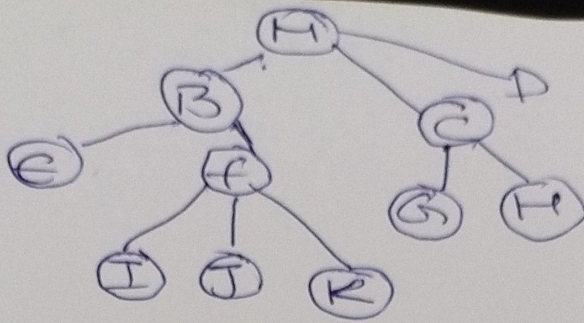


In order (LNR) = 4, 6, 3, 7, 2, 5, 1

Post order (LRN): 4, 3, 6, 2, 1, 5, 7

Pre order (NLR): 7, 6, 4, 3, 5, 2, 1

10)



Pre order: A, B, E, F, I, J, K, L, G, H, D

Postorder: - E, I, J, K, F, B, G, H, C, D, A

level order: A, B, C, D, E, F, G, H, I, J, K